J Autism Dev Disord (2008) 38:395–400
DOI 10.1007/s10803-007-0392-2

BRIEF REPORT

Enhancing the Conversation Skills of a Boy with Asperger's Disorder through Social Stories™ and Video Modeling

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Published online: 2 June 2007
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Abstract This study combined Social Stories with video modeling in an effort to enhance the conversation skills of a boy with Asperger's Disorder. Treatment consisted of two components: (a) observation of video taped Social Stories that included two adults modeling targeted conversational skills and (b) 5-min social interactions. A multiple baseline design across behaviors was used to evaluate the intervention and an increase in 2 out of 3 targeted conversation skills occurred. In addition, generalized behavior changes were observed. These findings provide support for including Social Stories as part of a video treatment package in teaching complex social interaction behaviors to young children with Asperger's Disorder.

Keywords Asperger's Disorder · Social stories · Social interactions · Video modeling

Introduction

Profound impairments in social interaction are among the defining features of Asperger's Disorder. These deficits may be due, in part, to difficulties in understanding the unspoken yet complex rules that govern social behavior. To date, several strategies have been developed that target these core deficits (e.g., Oke & Schreibman, 1990; Zanolli, Daggett, & Adams, 1996). Some of these strategies such as Social Stories™ and video modeling utilize visual cues to teach children with autism spectrum disorders appropriate social skills (e.g., Charlop & Milstein, 1989; Gray, 2000).

Social Stories are written short stories that provide the child with an autism spectrum disorder accurate social information about an activity or event, a description of the possible reactions of others, and direction as to the responses he or she is expected to provide (Gray, 2000) in a given social situation. They generally consist of several pages compiled into a book-like format and are written from the child's perspective. Current research has shown them effective for decreasing disruptive behavior (Kuttler, Myles, & Carlson, 1998; Scattone, Wilczynski, Edwards, & Rabian, 2002), enhancing greetings (Swaggart et al., 1995), and teaching appropriate play (Barry & Burlew, 2004), for example. They have also been used to foster conversation skills without the combination of another intervention (Scattone, Tingstrom, & Wilczynski, 2006) and when combined with video feedback (Thiemann & Goldstein, 2001).

Video modeling has also been found effective in enhancing conversation skills for children with autism. For example, Charlop and Milstein (1989) used video modeling to teach conversational skills to three boys with autism. The children observed two adults on videotape talking about specific toys and then were instructed to ''do the same.'' All three children improved their conversational speech, variations in their responses were observed, and skills generalized to different settings and conversational partners.

Given the complexities of the impairments in social interaction demonstrated by these children, strategies that incorporate several components of visually cued instruction may enhance the effectiveness of the Social Story intervention. Thus, the present study examined the impact of Social Stories combined with video modeling on the conversation skills of a boy with Asperger's Disorder.

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Method

Participant

Matthew, a 9-year old boy with Asperger's Disorder, was referred to a clinic in the South by his mother for difficulties with social interactions. Matthew demonstrated poor eye contact and often engaged in one-sided perseverative speech without an awareness of whether the interaction partner was either engaged in the conversation or wanted to comment. His mother and teachers often attempted to teach him appropriate social behaviors; however, he failed to acquire the basic skills necessary to maintain reciprocal conversation.

Matthew's IQ, as measured by the Kaufman Brief Intelligence Test yielded a Vocabulary score of 117, a Matrices score of 99, and a Composite IQ of 109 (KBIT; Kaufman & Kaufman, 1990). His Basic Reading score on the Wechsler Individual Achievement Test was 126 and his Numerical Operations score was 114 (WIAT; Wechsler, 1992). Matthew was diagnosed according to the DSM-IV-TR (2000) criteria for Asperger's Disorder by the author, who is a licensed psychologist with 5 years experience in the field of autism.

Setting

Treatment sessions were held at a medical center in a small room and videotaped. There were 24 sessions over a period of 15 weeks.

Design

A multiple baseline design across behaviors was used to assess changes in target skills (Hayes, Barlow, & Nelson-Gray, 1999). Initially, baseline data were collected simultaneously for target behaviors. The Stories were introduced in a staggered fashion beginning with eye contact then systematically applied to smiling, and then initiations. A pre-intervention probe was taken to assess Matthew's interactions with his peers at school and a generalization probe was also taken at Matthew's school at the conclusion of the study.

Dependent Variables and Data Collection

Target skills included eye contact, smiling, and initiations. Eye contact was defined as looking at the interaction partner for 3 s or more. Smiling was defined as grinning or laughing. Initiations were defined as any unprompted question or comment made by Matthew that was directed to the interaction partner. For example, ''You look nice today'' or ''What are your plans this weekend?''

Data on social interactions were collected during 5-min observations 1–2 times per week. A partial-interval recording procedure was used to record target skills during the observations. Researchers were cued by an audiotape every 10 s to record the occurrence of the target behaviors. Data were scored from videotapes. The video recorder was located close enough to the participant to observe his face clearly; however, it did not appear to interfere with the interactions.

Interobserver Agreement

Interobserver agreement was measured by assessing level of agreement and disagreement over the course of the study. Interobserver agreement was defined as occasions when both observers agreed that a behavior either occurred or did not occur. Disagreements were scored if the observers did not agree on the type of social interaction that occurred or if one observer did not record a behavior. Percentages of interobserver agreement were computed by dividing the number of agreements by the total number of agreements plus disagreements and multiplying by 100.

Interobserver agreement was assessed 11 times representing 46% of the observations. The mean level of interobserver agreement for eye contact was 98% (range = 97–100%), for smiling was 98% (range = 96–100%), and initiations were 99% (range = 96–100%).

Materials

Three Social Stories were developed according to Gray's basic social story format and each described the target skill(s) and suggested responses. The first story focused on eye contact, the second story on eye contact and smiling, and the third story focused on all three target behaviors: eye contact, smiling, and initiations. Each Social Story was compiled into a booklet with typed white pages mounted on black paper. The Social Stories were 6, 7, and 10 pages respectively and the wording of each page was shown in the video and narrated by an adult. After the Social Story was narrated, two adults modeled the target skill (e.g., eye contact) during a 5-min video taped conversation.

Each videotaped Social Story was initially viewed in the clinic setting with the first author. After Matthew viewed the video, his comprehension was assessed via predetermined questions the first time each new Social Story was introduced. Matthew answered each Story's questions with 100% accuracy the first time they were asked. Thereafter, Matthew viewed the video once at home with mother's supervision each evening and again in the clinic just before data collection. Intervention data were collected in the same manner as during baseline.

Procedure

Throughout baseline and intervention phases 5-min social interaction sessions were conducted 1–2 times per week and videotaped using a recorder located near the participant. Before the Social Stories were introduced, a researcher collected baseline data of Matthew having a conversation with an adult in the clinic. One of five adults served as interaction partners over the course of the study. During each observation the occurrence of eye contact, smiling, and initiations were recorded.

A baseline data probe was also taken at Matthew's school during lunchtime and eye contact, smiling, and initiations were recorded in the same manner as clinic baseline. However, typical peers who were sitting with Matthew at lunchtime served as interaction partners rather than adults. They were not instructed as to the purpose of the study. School probes were not videotaped.

Treatment Integrity

Matthew's mother completed a daily checklist and recorded whether or not he watched the video that evening. Treatment integrity was stated as a percentage and calculated by dividing the number of days the participant viewed the Social Story by the number of total days in the intervention phase multiplied by 100. According to parent report, treatment integrity was 95%.

Data Analysis

The effectiveness of the intervention on the participant's social interactions was assessed by measuring the occurrence of intervals of eye contact, smiling, and initiations during 5-min observations across baseline and intervention phases. Social interactions for each target skill were graphed as a percentage of intervals (see Fig. 1). Changes in level, variability, and trend were inspected visually during baseline and intervention (Hayes, Barlow, & Nelson-Gray, 1999; Kazdin, 1982).

Results

The mean level of eye contact during baseline was 6% of intervals (range = 0–13%) and 97% (range = 67–100%) during intervention. The mean level of smiling during baseline was .6% (range = 0–3%) and was variable during intervention with a mean level of 7% (range = 0–40%). Initiations reached a mean level of .8% of intervals (range = 0–7%) during baseline and 33% (range = 20–52%) during intervention (See Fig. 1).

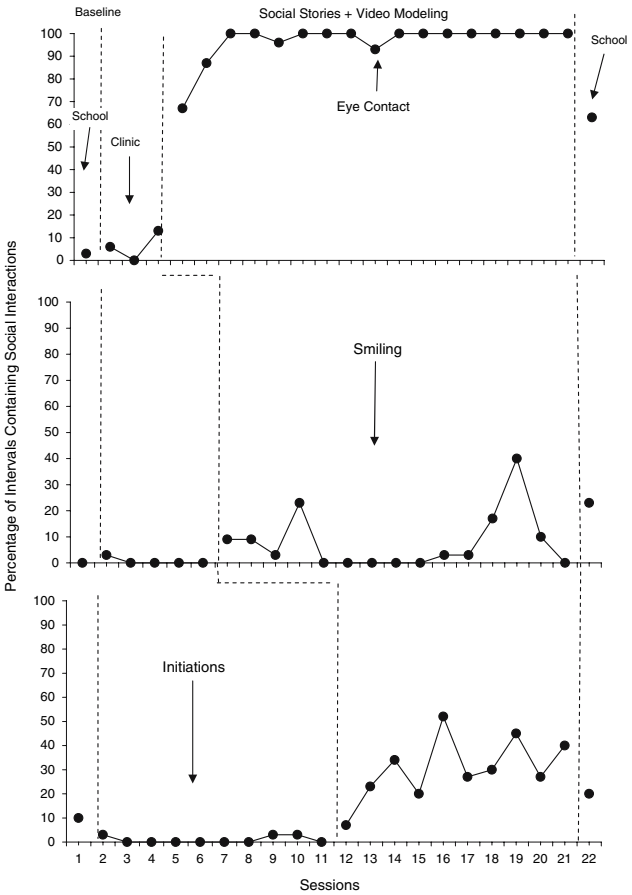


Fig. 1 Percentage of intervals containing Social Interactions

Baseline data from the probe taken at Matthew's school during lunchtime with peers reached a mean level of 3% of intervals for eye contact, smiling for 0 intervals, and reciprocal social interactions for 10% of intervals. Post intervention data also taken at school directly after the last data point was collected in the clinic demonstrated improvement over both school and clinic baseline as Matthew engaged in eye contact for 63% of intervals, smiling for 23% of intervals, and reciprocal interactions for 20% of intervals.

To assess intervention effectiveness, percentage of nonoverlapping data (PND) for appropriate social interactions between phases was computed by dividing the number of data points in intervention that did not overlap with data points in baseline and then by dividing that figure by the total number of points in the intervention phase. PND were calculated for each of Matthew's target behaviors. PND for eye contact was 100%, PND for smiling was 32%, and PND for reciprocal interactions was 100%. Scruggs, Mastropieri, Cooke, and Escobar (1986) provided a general scale to interpret PND. Specifically they suggest a nonoverlapping data above 90% indicates a highly effective outcome, 70–90% represents a fair outcome, 50–70% represents a

questionable outcome, and below 50% suggests an unreliable treatment. Thus, a PND of 100% for eye contact and reciprocal interactions was a considered a highly effective outcome, whereas, 32% between baseline and intervention phases for smiling was considered unacceptable.

An effect size was also calculated to measure the magnitude of treatment effect for each phase. Cohen (1988) defined effect size as large if .9 and above, medium if .3 to .5, and small if below .2. The effect size for eye contact was .9 suggesting a large effect. The effect size was .4 for smiling indicating a medium effect. Finally, the effect size for initiations was .8 indicating a large effect.

Matthew's mother completed the Intervention Rating Profile (IRP-15; Martens, Witt, Elliott, & Darveaux, 1985) at the conclusion of the study to assess her acceptability of the social story intervention. The IRP-15 is a 15-item scale used to evaluate teacher's acceptability of a given intervention. Scores on the IRP-15 can range from 15 to 90, with higher scores indicating greater acceptance. Scores above 52.50 are considered to indicate that an intervention is acceptable (Von Brock & Elliott, 1987). Her responses on the IRP-15 were 89 falling well within the acceptable range.

Discussion

Social Stories combined with video modeling were effective in increasing the conversation skills of a boy with Asperger's Disorder for two of the three target behaviors, although the level of improvement for each skill differed. When treatment was introduced, Matthew immediately demonstrated improved eye contact, which remained high and steady throughout treatment and the school probe.

Smiling was a difficult skill for Matthew to master and was variable throughout treatment. During baseline, smiling averaged .6% of intervals and 7% of intervals after treatment indicating no appreciable improvement occurred. However, smiling during the generalization probe taken at school improved moderately from clinic baseline levels but was inconsistent throughout the study. It should be noted that at the conclusion of the study, smiling continued to be targeted in a variety of ways (e.g., using a mirror, verbal prompts) during the remainder of his treatment in the clinic, and remained a challenging skill for Matthew to master.

Eye contact appeared to be increasing slightly by the fourth baseline data point, just before the intervention was implemented, therefore, results should be interpreted with caution. In addition, without maintenance data it remains undetermined whether or not eye contact remained strong or required continued intervention in Matthew's continued social interactions.

Initiations also improved from an average of .8% of intervals during baseline to an average of 33% of intervals during treatment. However, initiations did not increase much from baseline probe at school to post intervention probe at school (23% of intervals).

Matthew demonstrated some generalized treatment effects such as increased eye contact, smiling, and reciprocal social interactions with peers at school. For example, during the baseline probe, he demonstrated low levels of all three behaviors; however, during a generalization probe, eye contact, smiling, and initiations were lower than clinic sessions but still higher than baseline. Although Matthew did not make many initiated comments, anecdotally he was observed to appear engaged in conversation with his peers by nodding his head in agreement and smiling which were behaviors not observed during the baseline school probe or in the clinic.

In addition, Matthew was observed to have talked about several topics that were mentioned in the story both in the clinic and at school. For example, he asked about favorite foods, pets, and activities for the weekend, topics that were mentioned in the Social Story video. He was also observed, although less frequently, to give the interaction partner a compliment (e.g., ''You look nice today.''). Twice he asked about things that were not mentioned in the Social Story video (e.g., favorite colors). However, it should be noted that there was no appreciable difference between initiations between pre and post intervention at school and most of increase was due to scripted topics directly from the video.

Although Matthew viewed a Social Story video each night for the duration of the study (i.e., 3 months), mother reported that it was difficult to maintain his interest in viewing the video towards the conclusion of the study, especially after he had mastered two of the skills fairly quickly. Thus, to maintain interest, it may be helpful to use multiple Stories with similar content. Another alternative may be to add a reinforcement component to assist with treatment compliance.

Because the intervention involved two components (i.e., Social Stories and video modeling) that were not systematically introduced, it is impossible to determine how the individual components contributed to the effectiveness of the Social Stories. Given that social stories and video modeling have both been shown to be effective treatments, and both are often used in schools and clinics, it isn't clear that combining both interventions provides a substantial contribution to the literature. However, investigating whether video modeling plus Social Stories are more effective than the either the Social Story or video alone would be a worthwhile exploration.

Because only one pre intervention and one generalization probe was taken at the conclusion of the study, the data obtained from single probes must be viewed with extreme caution.

Matthew was referred to the clinic for difficulties in his peer relationships, thus future research should include having those peers model target behaviors in the video. In addition, viewing the video taped conversation to allow for self-evaluation as in the Theimann and Goldstein (2001) study may also be a worthwhile variable to explore. For example, smiling may have increased if Matthew had the opportunity to view the conversation and then evaluate his own performance.

This study is clearly limited by having only one student participate in the study. Therefore, future research should also examine the effects of this intervention in a multiple baseline across participants. In addition, 5-min observations are relatively short, but for this participant who had difficulty sustaining even short interactions, it was thought to be a good starting point for him. Future research could include lengthening the observations from 5 min to 10 min or longer over a period of time to see if the participant can sustain the skills in acquisition.

In conclusion, Social Stories plus video modeling may be an effective intervention for some student's with Asperger's Disorder; however, the exact role that video modeling plays in conjunction with Social Stories warrants further exploration.

Appendix

Social Stories for Eye Contact, Smiling, & Initiations

Looking and Talking

When I come to the clinic, I see lots of people. Some are nurses. Some are doctors. Some are kids! Usually someone talk to me. When someone talks tome, I will try to look at part of their face. Most people like it when I look at their face. Looking at their face makes them feel good. This lets them know I'm listening. If I look at them when I am talking, they will like this a lot. They will think I'm a nice boy I will try to look at their face most of the time we are talking.

Comprehension Questions:

1. Who do I see at the clinic?
2. What should I try to do if someone talks to me?
3. Do most people like it when I look at them while we talk?

Smiling

I like coming to the clinic. Most of the time, the people at the clinic say ''hello'' and ask me how I'm doing. Sometimes someone will sit and talk with me. It could be Mary, Susie, Annie, Maria or it could be someone else. When someone talks to me, I will try to look at part of their face. This way they know I'm listening. I will also try to smile when I look at them. If I smile while I'm looking at their face, they will like this a lot. Smiling will make them feel good to be around me. It will let them know I'm listening and that I'm enjoying talking to them. I will try to look and smile wen someone is talking to me.

Comprehension Questions:

1. Who usually talks to me in the clinic?
2. What someone sits and talks with me in the clinic what should I do?
3. How do people feel when I smile at them while we talk?

Taking Turns Talking

When people in the clinic see me, they usually say, ''Hi Matthew!'' When we talk, I will try to look at part of their face and I will try to smile! They may say, ''How are you Matthew?'' or ''How was school?'' After someone asks me a question, I will try to answer them in ONLY 1 or 2 sentences. Then I will try to give them a TURN to talk! I will try to *ask them a question about themselves*. Or…I can say something nice to them! This is called taking turns asking questions. They may ask a question, then I ask a question, and we go back and forth each taking turns.

I can ask: ''How are you?''
''What are you going to do this weekend?''
''Did you see any movies?''
''Do you have pets? What kind?''
I can also say nice things like:
''You look nice today!'' or
''I really like your outfit'' or
''I really enjoy talking with you!''

If I look at the other person, smile, and ask how THEY are or I say nice things, this will let them know I like them! They will want to be my friend!

Comprehension Questions:

1. After someone asks me a question, what should I do?
2. What can I ask other people?
3. What nice can I say to other people?

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